ANASUYA ACHARYA

Assistant Professor at Aalborg University in Copenhagen

anasuyahirai@gmail.com | +45 52742305 | Copenhagen, Denmark
www.linkedin.com/in/anasuyaacharya | anasuyaacharya.github.io/

EXPERIENCE

Postdoctoral Researcher
Aarhus University
March 2025 – July 2026 | Aarhus, Denmark

Postdoctoral Researcher in secure multiparty computation (MPC) hosted by Prof. Claudio Orlandi in the Cryptography and Cybersecurity Section, Aarhus University.

Research Assistant
Information Security Research and Development Center (ISRDC)
July 2017 – July 2020 | Mumbai, India

A three year scholarship student and Research Assistant studying distributed ledger technologies (DLT) in ISRDC, IITB.

Research Intern
Indian Statistical Institute
December 2015 – January 2016 | Kolkata, India

Research intern in cryptography under Prof. Shubhamoy Maitra, studying and attempting cryptanalysis of the RC4 stream cipher.

SERVICE

Workshop Talks

- *Maliciously Secure SCALES Protocols* (2024). Cryptography in the Blockchain Era (CIBE) 2024, 23-27 June 2024, Bertinoro, Italy.
- *Maliciously Secure SCALES Protocols* (2024). Theory and Practice of Multi-Party Computation Workshop (TPMPC) 2024, 3-6 June 2024, TU Darmstadt, Germany.
- *SCALES: MPC with Small Clients and Larger Ephemeral Servers* (2023). ACE Conference (Algorand Centres of Excellence) 2023, 10-12 January 2023, Hotel Arts, Barcelona.
- *Maliciously Secure SCALES Protocols* (2022). Algorand Centres of Excellence (ACE) Seminar, Online, November 2022.
- *SCALES: MPC with Small Clients and Larger Ephemeral Servers* (2022). Theory and Practice of Multi-Party Computation Workshop (TPMPC) 2022, 7-10 June 2022, Aarhus University, Denmark.
- *CellTree: A new Paradigm for Distributed Data Repositories* (2019). Talk, 28 June 2019, IIT Bombay, Mumbai, India.

Organizing

- *Theory and Practice of Multi-Party Computation Workshop (TPMPC 2026)* (2026). 18-22 May 2026, Aarhus University, Aarhus, Denmark.
- *12th BIU Winter School on Cryptography - Advances in Secure Computation* (2022). 23-26 January 2022, Bar-Ilan University, Israel.
- *39th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science* (2019). 11–13 December 2019, IIT Bombay, Mumbai, India.
- *ACM India Grad Cohort 2018* (2018). 6-7 July 2018, IIT Bombay, India.

EDUCATION

Ph.D. in Computer Engineering
Bar-Ilan University
July 2025 | Ramat Gan, Israel

CGPA: 86.94 / 100

M.Tech. in Computer Science and Engineering (CSE)
IIT Bombay
July 2020 | Mumbai, India

CGPA: 9.54 / 10

B.Tech. in CSE
NIT Trichy
May 2017 | Tiruchirappalli, India

CGPA: 8.06 / 10

REFEREES

Prof. Claudio Orlandi
Aarhus University

Aarhus, Denmark
orlandi@cs.au.dk
+45-21433249
https://users-cs.au.dk/orlandi/
Postdoc Host

Prof. Carmit Hazay
Bar-Ilan University

Ramat Gan, Israel
carmit.hazay@biu.ac.il
+972-3-738-4672
https://www.eng.biu.ac.il/hazay/
PhD Thesis Supervisor

Prof. Manoj M. Prabhakaran
IIT Bombay

Mumbai, India
mp@cse.iitb.ac.in
+91-22-2576-7709
www.cse.iitb.ac.in/~mp/
Masters Thesis Supervisor

PUBLICATIONS

H-index : 5 (according to Google Scholar)
ORCID : orcid.org/0000-0002-9111-5641
Following the Hardy-Littlewood rule, the list of authors in most of my publications is sorted alphabetically.

Conference Proceedings

- Acharya, Anasuya, Carmit Hazay, and Rahul Satish (2026). "The Landscape of Reusable Garbling". In: *SCN 2026, September 14-16, 2026*. Amalfi, Italy: Springer.
- Acharya, Anasuya, Pierre Meyer, Divya Ravi, and Rahul Satish (2026). "Sublinear-Communication layered MPC from HSS". in: *ASIACRYPT 2026, December 7-11, 2026*. Hong Kong, China: Springer.
- Acharya, Anasuya, Aditya Patankar, Arpita Patra, Divya Ravi, and Raghavendra Vernekar (2026). "Constant-round MPC protocols with Fall-back Security". In: *TCC 2026, November 10-13, 2026*. New York, USA: Springer.
- Acharya, Anasuya, Karen Azari, Mirza Ahad Baig, Dennis Hofheinz, and Chethan Kamath (2025). "Securely Instantiating 'Half Gates' Garbling in the Standard Model". In: *PKC 2025, May 12-15, 2025*. Røros, Norway: Springer, pp. 37–75.
- Acharya, Anasuya, Karen Azari, and Chethan Kamath (2025). "On the Adaptive Security of Free-XOR-Based Garbling Schemes in the Plain Model". In: *EUROCRYPT 2025, May 4-8, 2025*. Madrid, Spain: Springer, pp. 214–244.
- Acharya, Anasuya, Carmit Hazay, Vladimir Kolesnikov, and Manoj Prabhakaran (2025). "Towards Building Efficient SCALES Protocols". In: *ASIACRYPT 2025, December 8-12, 2025*. Melbourne, Australia: Springer.
- Acharya, Anasuya, Carmit Hazay, and Muthuramakrishnan Venkitasubramaniam (2025). "On Achieving "Best-in-the-Multiverse" MPC". in: *TCC 2025, December 1-5, 2025*. Aarhus, Denmark: Springer.
- Acharya, Anasuya, Carmit Hazay, Vladimir Kolesnikov, and Manoj Prabhakaran (2024). "Malicious Security for SCALES - Outsourced Computation with Ephemeral Servers". In: *CRYPTO 2024, August 18-22, 2024*. Santa Barbara, USA: Springer, pp. 3–38.
- Acharya, Anasuya, Tomer Ashur, Efrat Cohen, Carmit Hazay, and Avishay Yanai (2023). "A New Approach to Garbled Circuits". In: *ACNS 2023, June 19-22*. Kyoto, Japan: Springer, pp. 611–641.
- Acharya, Anasuya, Carmit Hazay, Oxana Poburinnaya, and Muthuramakrishnan Venkitasubramaniam (2023). "Best of Both Worlds - Revisiting the Spymasters Double Agent Problem". In: *CRYPTO 2023, August 20-24, 2023*. Santa Barbara, USA: Springer, pp. 328–359.
- Acharya, Anasuya, Carmit Hazay, Vladimir Kolesnikov, and Manoj Prabhakaran (2022). "SCALES - MPC with Small Clients and Larger Ephemeral Servers". In: *TCC 2022, November 7-10, 2022*. Chicago, USA: Springer, pp. 502–531.
- Prabhakaran, Manoj M., Anasuya Acharya, and Akash Trehan (2019). "An Introduction to the CellTree Paradigm". In: *ICISS 2019, Volume 11952 LNCS, 16-20 December 2019*. Hyderabad, India: Springer.
- Sanyashi, Tikaram, Anasuya Acharya, and Bernard Menezes (2019). "Plaintext Recovery Attacks and their Mitigation in an Application Specific SHE Scheme". In: *PDCAT 2019, 5-7 December 2019*. Gold Coast, Australia: IEEE Xplorer.
- Patil, Vishwas, Anasuya Acharya, and R. K. Shyamasundar (2018). "Landcoin: A Land Management System using Litecoin Blockchain Protocol". In: *SDLT3 2018, 12 November 2018*. Gold Coast, Australia.

REVIEWING

PC Member

TCC '26 **ASIACRYPT '26**

Sub-reviewing

CRYPTO '22, '24, '25, '26
EUROCRYPT '22, '24, '25
ASIACRYPT '25
PKC '26
ITC '25
TCC '22, '24
PODC '26

ACHIEVEMENTS

GATE (Graduate Aptitude Test for Engineering)
Score of 791 and AIR (All India Rank) 231 in the GATE Computer Science and Information Technology paper in February 2017

AP (Advanced Placement Exam)
Scored 3/5 in Physics C Mechanics, Computer Science A, Calculus AB in May 2012

AAT (National Award of Academic Aptitude and Achievement Test)
Scored 99 percentile in quantitative aptitude in May 2007

SOFTWARE SKILL SET

LaTeX | GIT | Linux(Ubuntu)

Java | C | C++ | SEBOL | JavaScript | Python | PHP | MATLAB

LANGUAGES

English	●●●●●
Hindi	●●●●●
Bengali	●●●●○
Japanese	●●●●○
Mandarin	●●○○○
Bahasa Indonesia	●●○○○
Danish	●●○○○

EXTRA-CURRICULAR

JLPT N2	●●●●○
HSK 3	●●●●●
TEFL	●●●●●
BMC @ HMI	●●●○○

HOBBIES

Trekking | Learning Languages | Writing | Crash Coursing | Visual Arts